SRH Computer Science BSc

PlantBully

Smart Plant Health Monitor

Karim
Verzhinia
Yueqingchen
Hasan

Plants require specific environmental conditions (temperature, humidity, soil moisture, and light) to grow properly.
Manual monitoring is inefficient and inconsistent. When conditions fall outside optimal ranges, plant health and yield decline.

An IoT-based monitoring system that continuously tracks environmental parameters and notifies users can be applied by:

- Smart Greenhouses
- Commercial Farming
- Individual Plant Owners

What makes us stand out?

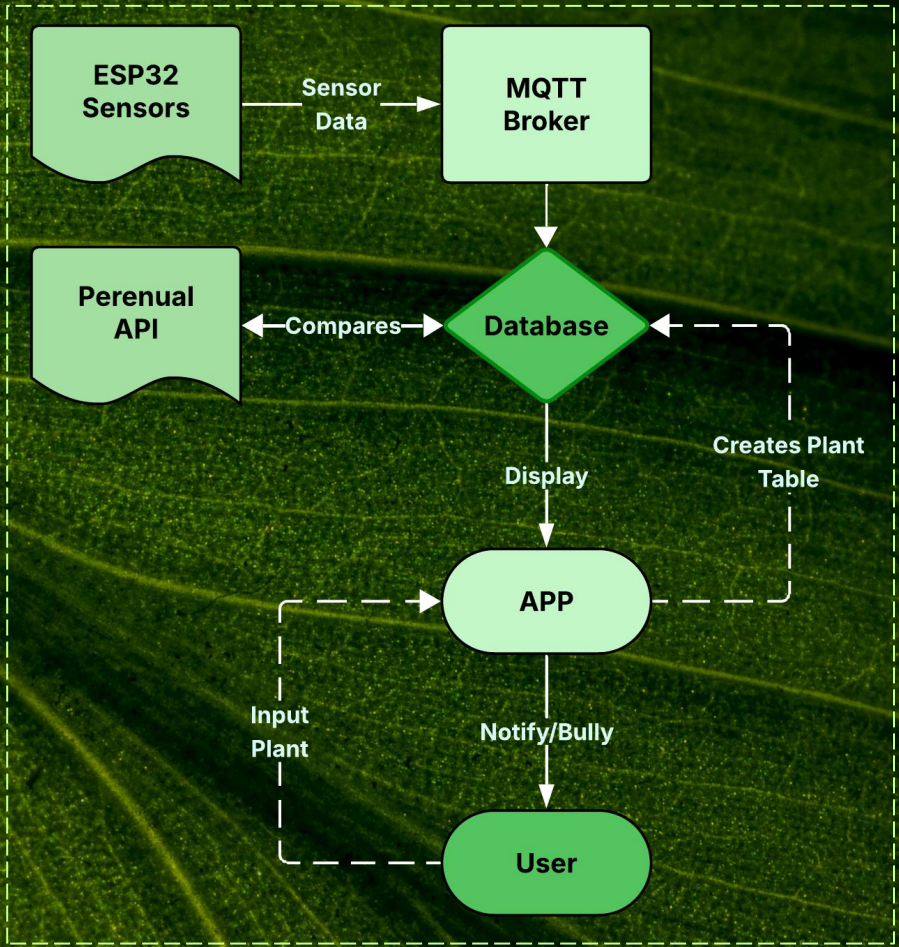
According to research, although they may seem passive, plants have their own complex sensory systems too, designed to respond to dangers or other changes in their environment. (Schetzer 2017)
In other words, plants have feelings too, and this project is the first app to aid them in expressing their feelings by bombarding their owner with text messages, addressing inclusivitiy for botanic lifeforms.

Softwares:

- Perenual API
- SQLite
- Flutter
- Visual Studio Code

Hardwares:

- ESP32-WROOM-D
- Photo Resistor
- DHT11 Sensor
- Capacitive Soil Sensor



ChatGPT was used to debug the code

1. Schetzer, A. (2017, April 17). Plants have feelings too. Pursuit. https://pursuit.unimelb.edu.au/articles/plants-have-feelings-too